Eduardo Hafemann

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Education

PhD in Mathematics **10/2023 – 09/2026**
Universität Hamburg (UHH)
Supervisor: Prof. Dr. Melanie Graf.
Co-supervisor: Prof. Dr. Eleni-Alexandra Kontou (King's College London).
Thesis: Approximation Techniques and Weakened Energy Conditions in Mathematical Relativity.

Msc Mathematics **08/2021 – 08/2023**
Federal University of Santa Catarina - Florianópolis (UFSC)
Supervisor: Prof. Dr. Ivan Pontual Costa e Silva.
Dissertation: Geometry and Topology of Black Hole Horizons (RI UFSC).

Bsc Chemical Engineering **08/2016 – 08/2021**
Federal University of Santa Catarina - Florianópolis (UFSC)
Award: Highest GPA in Chemical Engineering, Graduating Class of August 2021.

Papers in Mathematical Physics

Preprint

[1] E. Hafemann, "A low-regularity Riemannian positive mass theorem for non-spin manifolds with distributional curvature", 2026, arXiv:2602.03451 `[math.DG]`

[2] M. Graf and E. Hafemann, "Coordinate isoperimetric mass: density and positive mass theorem for distributional scalar curvature", 2026, (in preparation)

[3] E. Hafemann, E.-A. Kontou, and J. Procter, "Positive energy theorems and quantum energy inequalities", 2026, (in preparation)

Published

[4] E. Hafemann and E.-A. Kontou, "Penrose inequality for integral energy conditions", Classical and Quantum Gravity **42**, 195016 (2025)

[5] M. Calisti, M. Graf, E. Hafemann, M. Kunzinger, and R. Steinbauer, "Hawking's singularity theorem for Lipschitz Lorentzian metrics", Commun. Math. Phys. **406**, 207 (2025)

Papers in Applied Fields

Inverse Problems

[6] A. De Cezaro, E. Hafemann, and A. Leitão, "A piecewise constant levelset approach for semi-blind deconvolution: application to barcode decoding", 2026, Preprint

[7] A. De Cezaro, E. Hafemann, A. Leitão, and A. Osses, "A regularization method based on level-sets for the problem of crack detection from electrical measurements", Inverse Problems **39**, 035009 (2023)

[8] R. Filippozzi, E. Hafemann, J. C. Rabelo, F. Margotti, and A. Leitão, "A range-relaxed criteria for choosing the Lagrange multipliers in the Levenberg–Marquardt–Kaczmarz method for solving systems of non-linear ill-posed equations: Application to EIT-CEM with real data", Journal of Inverse and Ill-posed Problems **31**, 267–292 (2023)

[9] F. Margotti and E. Hafemann, "Range-relaxed strategy applied to the Levenberg–Marquardt method with uniformly convex penalization term in Banach spaces", Inverse Problems **38**, 095001 (2022)

Particle Physics

[10] B. C. Backes, E. Hafemann, I. Marzola, and D. P. Menezes, "Density-dependent quark mass model revisited: thermodynamic consistency, stability windows and stellar properties", Journal of Physics G: Nuclear and Particle Physics **48**, 055104 (2021)

Chemical Engineering

[11] T. Neitzel, C. S. Lima, E. Hafemann, D. A. A. Paixão, J. M. Junior, G. F. Persinoti, L. V. dos Santos, and J. L. Ienczak, "RNA-seq based transcriptomic analysis of the non-conventional yeast Spathaspora passalidarum during Melle-boinot cell recycle in xylose-glucose mixtures", Renewable Energy **201**, 486–498 (2022)

[12] E. Hafemann, R. Battisti, D. Bresolin, C. Marangoni, and R. A. F. Machado, "Enhancing chlorine-free purification routes of rice husk biomass waste to obtain cellulose nanocrystals", Waste and Biomass Valorization **11**, 6595–6611 (2020)

[13] E. Hafemann, R. Battisti, C. Marangoni, and R. A. Machado, "Valorization of royal palm tree agroindustrial waste by isolating cellulose nanocrystals", Carbohydrate Polymers **218**, 188–198 (2019)

[14] R. Battisti, E. Hafemann, C. A. Claumann, R. A. F. Machado, and C. Marangoni, "Synthesis and characterization of cellulose acetate from royal palm tree agroindustrial waste", Polymer Engineering and Science **59**, 891–898 (2018)

Books

Inverse Problems

[15] F. Margotti, E. Hafemann, and L. M. Santana, *Implementação computacional da tomografia por impedância elétrica (Computational implementation of electrical impedance tomography)*, ISBN 978-85-244-0535-8 (Editora do IMPA, 2023)

Talks

Invited Talks

Geometry Seminar, *Radboud University* [1] — May 2026
Differential Geometry Seminar, *Federal Univ. of Santa Catarina* [1] — Mar 2026
Gravity Journal Club, *King's College London* [1] — Dec 2025
Mathematical Relativity Seminar, *Universität Hamburg* [4] — Apr 2025
Mathematical Relativity Seminar, *Universität Hamburg* [1] — Dec 2024

Contributed Talks

15th Central European Relativity Seminar, *Radboud University* [1] — Jan 2025
34th Brazilian Mathematics Colloquium, IMPA [9, 15] — Jul 2023

Research Stay & Training

Research Stay

King's College London, Prof. E.-A. Kontou, 1 week — Dec 2025
King's College London, Prof. E.-A. Kontou, 1 week — May 2024
Universität Wien, Profs. M. Kunzinger and R. Steinbauer, 1 week — Dec 2023

Workshop, Masterclass and Meeting

Masterclass, *Geometrical Aspects of Mathematical Relativity*, University of Copenhagen. — Jun 2025

Workshop, *Introductory Workshop – New Frontiers in Curvature*, SLMath, Berkeley, CA (Travel support of $1,000). — Aug 2024

Meeting, *14th Central European Relativity Seminar*, University of Tübingen. — Feb 2024

Teaching Experience

Theory of Distributions Teaching Assistant — **10/2024 – 03/2025**
Responsible for creating exercise sheets and exercise classes.

Differential Geometry Teaching Assistant — **08/2022 – 12/2022**
Assisted undergraduate mathematics students with homework and review sessions.

Language & Skills

Languages
Brazilian-Portuguese (Native), English (Fluent), German (A1).

Computer Skills
Python, FEniCS, R, MATLAB, Docker, Git, Django, HTML, CSS, JavaScript.

References

Prof. Dr. Melanie Graf
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Prof. Dr. Eleni-Alexandra Kontou
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